

Exploring Object-Oriented Principles in Deep Learning Frameworks: PyTorch vs. TensorFlow

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Abstract

All code and writeup is stored in this repo
https://github.com/XUJIAZHANG2002/CSCI5448_project

Topic

PyTorch and TensorFlow are currently two leading deep learning frameworks. This project will examine how these frameworks apply object-oriented (OO) principles in their design. To be more specific, we will identify where core OO concepts such as encapsulation, inheritance, abstraction, polymorphism, and composition appear in each framework. In particular, we would look deeply into the design of `nn.Module` in PyTorch and `tf.keras.Model` in TensorFlow, as well as the use of common design patterns including Composite, Strategy, and Observer.

Example code snippets and runnable tests will be provided to illustrate how OO abstractions manifest in model building, extension, training and deployment. Additionally, we would perform a comparative study between PyTorch and TensorFlow, by implementing them to solve same tasks, such as CNN-based image classification and attention-based language generation.

Motivation

The motivation for this topic comes from the central role that PyTorch and TensorFlow play in modern AI developments. Both frameworks are widely used in academia and industry to drive the rapid growth of AI applications. In this project, we want to learn both libraries deeper by analyzing how each implementation follows OO principles to provide a user-friendly interface.

This analysis is significant to the wider computer science and software engineering community because it bridges traditional OO design education with cutting-edge machine learning software. We hope this will motivate the communication between the software engineering community and AI community to build better AI libraries in the future.